

Jeroen Van Goey

Staff Research Engineer in BioAI building generative models for biology

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Summary

Machine learning research engineer working at the intersection of AI and science, bridging scientific ML, biological domain knowledge, production-quality software, and team leadership. I build generative models and take them from prototype to deployment, with a focus on *de novo* peptide sequencing.

8 publications, including papers published in *Nature Machine Intelligence* and *Nature Communications*.

Currently looking for Staff/Principal Research Engineer roles in AI for science. Based in Cape Town, open to remote-first roles now and hybrid/on-site roles in Europe or the USA from 2027.

Experience

InstaDeep, Staff Research Engineer · BioAI Lead

Cape Town, South Africa
Aug 2022 – present
4 years 2 months

- **Tech lead and hiring manager** for two cross-disciplinary **ML research teams** of about 5 to 8 people: *de novo* peptide sequencing (with the Technical University of Denmark) and signal-peptide design for secretion efficiency (with BioNTech).
- Delivered a **client collaboration with Syngenta**, applying **genomic language models** (AgroNT, a Nucleotide Transformer trained on ~10.5 million genomic sequences spanning trillions of base pairs across 48 plant species) to accelerate crop trait research.
- Set **research direction** and take models **from research to production** (InstaNovo is available to commercial customers on **DeepChain**), balancing scientific rigour with reliable engineering.
- **Lead the BioAI team** of InstaDeep's Cape Town office and serve as the office's **site manager** (an office of about 25 people); hire and grow the BioAI teams across the Cape Town and Kigali offices.
- Directed the team's **ML engineering foundations**: scalable, Transformer-based libraries (Python / PyTorch) for large-scale training on InstaDeep's Kyber (~500 PFLOPs) and EuroHPC's MareNostrum 5 (260 PFLOPs) supercomputers, and the cloud (AWS, GCP).

Barco, Senior Software Development Engineer, Machine Learning

Kortrijk, Belgium
Feb 2020 – Aug 2022
2 years 7 months

- Built a TensorFlow Extended production pipeline (orchestrated with Apache Airflow) training deep-learning models on multispectral images to detect and classify melanoma skin cancers for Demetra, Barco's dermatology imaging device.
- Engineered the pipeline with **end-to-end data and model lineage tracking** to deliver the traceability and reproducibility that **regulated medical industries (FDA approval)** require.
- Worked across the research, cloud-backend and mobile/web frontend teams.

Bayer Crop Science, then BASF, Bioinformatics Researcher, Manager of the Python & R Platforms

Zwijnaarde, Belgium
Feb 2018 – Jan 2020
2 years

- Owned the **Python/R data-analysis platform** used by **480 researchers and data scientists**: technical support, proactive monitoring and root-cause analysis, **training** (NumPy, pandas, BioPython) and **mentoring** across the company's internal research community.
- The Zwijnaarde site was divested from Bayer Crop Science to BASF as part of Bayer's acquisition of Monsanto; I continued in the same role across the transition.

Applied Maths (acquired by bioMérieux), Bioinformatics Software Developer

Sint-Martens-Latem, Belgium
Sept 2011 – Jan 2018
6 years 5 months

- Worked on BioNumerics, a bioinformatics suite for integrated analysis of biological data, writing **custom Python scripts and extensions** for clients in the clinical, academic and government sectors.

Selected publications

InstaNovo enables diffusion-powered *de novo* peptide sequencing in large-scale proteomics experiments

Nature Machine Intelligence
2025

K. Eloff, K. Kalogeropoulos, ..., **J. Van Goey**, ..., T. P. Jenkins

Trained on ~63 million spectra; top-ranked in an independent benchmark of 17 *de novo* tools across 83 datasets; Altmetric attention score 100+ and 120+ GitHub stars.

[10.1038/s42256-025-01019-5](https://doi.org/10.1038/s42256-025-01019-5)

InstaNovo-P: a *de novo* peptide sequencing model for phosphoproteomics

Nature Communications
2026

J. Lauridsen, V. Canbay, ..., **J. Van Goey**, ..., K. Kalogeropoulos

[10.1038/s41467-026-75138-x](https://doi.org/10.1038/s41467-026-75138-x)

Learning from tandem mass spectra at scale with a self-supervised foundation model for proteomics M. Nieuwoudt, M. Reverenna, ..., J. Van Goey , K. Kalogeropoulos 10.64898/2026.09.03.747733	bioRxiv (preprint) 2026
De novo peptide sequencing rescoring and FDR estimation with Winnow A. Mabona, J. Daniel, ..., J. Van Goey , K. Kalogeropoulos 10.48550/arXiv.2509.24952	arXiv 2025
A living proteomics benchmark for comprehensive evaluation of deep learning-based de novo peptide sequencing tools M. Pominova, J. Van Goey , C. Adams, ..., W. Bittremieux 10.6084/m9.figshare.31680883.v1	Registered report (preprint) 2026
Generalizable direct protein sequencing with InstaNexus M. Reverenna, M. Wennekers Nielsen, ..., J. Van Goey , ..., K. Kalogeropoulos 10.1016/j.mcpro.2026.101547	Mol. Cell. Proteomics 2026
A Framework for Database Search with AI Models in Mass Spectrometry-Based Proteomics K. Kalogeropoulos, J. Van Goey , T. P. Jenkins, K. M. Eloff 10.1021/acs.jproteome.5c01153	Journal of Proteome Research 2026

Education

HOWEST Hogeschool West-Vlaanderen , Microdegree Machine Learning & Deep Learning	Kortrijk, Belgium 2018 – 2019
Katholieke Universiteit Leuven , Bioinformatics (partial credits obtained)	Leuven, Belgium 2010 – 2012
University of Antwerp , M.Sc. Biology	Antwerp, Belgium 1997 – 2004
Universitat de Barcelona , Erasmus interuniversity exchange	Barcelona, Spain 2002
Møglestu videregående skole , AFS intercultural exchange program	Lillesand, Norway 1996 – 1997

Skills

Machine learning: Transformers, diffusion models, large language models (LLMs), supervised & self-supervised learning
Frameworks: PyTorch, TensorFlow / TFX, NumPy, pandas
MLOps: Kubernetes-based ML platforms (**Alchor**), Docker, CI/CD, workflow orchestration (Airflow, Dagster), experiment tracking (MLflow, Neptune, TensorBoard)
Cloud: Buckets, VMs, Cloud Run (AWS, GCP)
Scale: Distributed GPU training, HPC clusters (InstaDeep's Kyber and EuroHPC's MareNostrum 5)
Bioinformatics: BioPython, BLAST, ClustalW, Snakemake, BioNumerics
Programming languages: Python (primary), R
Spoken languages: Dutch (native), English, French & Norwegian (full professional), German & Spanish (notions)

Projects

Awesome De Novo Peptide Sequencing

A comprehensive, interactive map of the *de novo* peptide sequencing field: algorithms, post-processors, downstream applications, and adjacent tools, deep-learning and classical alike.

- Open source: github.com/BioGeek/awesome_de_novo_peptide_sequencing

Self-driving RC car

Turning a 1/16-scale RC car into an autonomous vehicle as a hands-on testbed for modern edge ML: training policies in simulation and deploying them to a Jetson Nano. An ongoing build, documented in a public progress log.

- Progress log: jeroen.vangoey.be/writing/jetson

SchaakStudieSpinsels 2

Digitised the second volume of my grandfather Ignace Vandecasteele's third book of chess endgame studies, more than three hundred studies, put the whole collection online, and made the chess boards interactive.

- Open source: github.com/BioGeek/SchaakStudieSpinsels2

Photography portfolio

A portfolio of portrait, wedding, maternity, boudoir, landscape and long-exposure photography.

Teaching & mentoring

AIMS South Africa MSc supervision

Co-supervised several AIMS South Africa Master's students at InstaDeep on de novo peptide sequencing and proteomics, including InstaNovo interpretability (sparse autoencoders), diffusion models for mass spectrometry, contrastive-predictive-coding encoder pretraining, glyco-peptide fine-tuning of InstaNovo, and AI-native proteomic database search via contrastive joint embeddings and a transformer-based scoring function.

AIMS South Africa, with InstaDeep
2025 – 2026

DTU PhD research stay

Mentored a Technical University of Denmark (DTU) PhD student during a 3-month research stay at InstaDeep Cape Town, developing a foundational model for proteomics with rigorous software-engineering practices, part of the ongoing DTU and InstaDeep collaboration.

InstaDeep, Cape Town, South Africa
2026

Machine Learning for Biology practical

Mentored the hands-on practical alongside InstaDeep and Google DeepMind colleagues.

Deep Learning Indaba, Ghana
2023

Scientific Python training & mentoring

Trained and mentored researchers and data scientists in scientific Python (NumPy, pandas, BioPython) as manager of the Python/R analysis platform.

BASF and Bayer Crop Science, Belgium
2018 – 2020

Hackathons (co-created, organized and judged)

2025: Machine Learning Interatomic Potentials Hackathon, IndabaX (Stellenbosch University, South Africa)

2024: Snakes and Sequences: Senegalese Serpent Venom Sequencing Hackathon, Deep Learning Indaba (Senegal)

2024: Desert Locust Breeding Ground Prediction Hackathon, IndabaX (Wits, Johannesburg)

2023: Unveiling Cassava's Secrets, Deep Learning Indaba (Ghana)